Done by: Samah Qaradeh

Notes: there is some codes that have no output, so I do not take screen shoots for them.

Write Parquet file using pandas:

homework.py 1 ×

C: > Users > USER > Desktop > iohealth > homework.py > ...

```python
import pandas as pd

# Sample DataFrame
data = {
    "Name": ["Alice", "Bob", "Charlie"],
    "Age": [25, 30, 35],
    "City": ["New York", "Los Angeles", "Chicago"]
}
df = pd.DataFrame(data)

# Write to Parquet file
df.to_parquet("data.parquet", engine="pyarrow", index=False)

print("Parquet file written successfully!")
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL

TERMINAL

```
PS C:\Users\USER\AppData\Local\Programs\Microsoft VS Code> & C:\Users\US
Parquet file written successfully!
PS C:\Users\USER\AppData\Local\Programs\Microsoft VS Code>
```

Read Parquet file using pandas:

homework.py 1 ×

C: > Users > USER > Desktop > iohealth > homework.py > ...

```python
import pandas as pd

# Read the Parquet file

df = pd.read_parquet("data.parquet", engine="pyarrow")

print("Data from Parquet file:")
print(df)
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL

TERMINAL

```
PS C:\Users\USER\AppData\Local\Programs\Microsoft VS Code> & C:\Users
omework.py
Data from Parquet file:
      Name  Age         City
0    Alice   25     New York
1      Bob   30  Los Angeles
2  Charlie   35      Chicago
PS C:\Users\USER\AppData\Local\Programs\Microsoft VS Code>
```

Write Parquet files using PyArrow

homework.py 1 ×

C: > Users > USER > Desktop > iohealth > homework.py > ...

```python
import pandas as pd
import pyarrow as pa
import pyarrow.parquet as pq

# Sample data
df = pd.DataFrame({
    "Name": ["Jacob", "Lauren", "Oliver"],
    "Age": [25, 30, 35],
    "City": ["New York", "Los Angeles", "Chicago"]
})

# Convert to a PyArrow table
table = pa.Table.from_pandas(df)

# Write to Parquet file
pq.write_table(table, "data.parquet")

print("Parquet file written successfully!")
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL

TERMINAL

```
PS C:\Users\USER\AppData\Local\Programs\Microsoft VS Code> & C:\Use
omework.py
Parquet file written successfully!
PS C:\Users\USER\AppData\Local\Programs\Microsoft VS Code>
```

Read Parquet files using PyArrow :

homework.py 1

C: > Users > USER > Desktop > iohealth > homework.py > ...

```python
import pyarrow.parquet as pq

# Read the Parquet file
table = pq.read_table("data.parquet")

# Convert to a pandas DataFrame
df = table.to_pandas()

print("Data from Parquet file:")
print(df)
```

CHAT

PORTS

TERMINAL

```
PS C:\Users\USER\AppData\Local\Programs\Mi
c:/Users/USER/Desktop/iohealth/homework.py
Data from Parquet file:
     Name  Age         City
0   Jacob   25     New York
1  Lauren   30  Los Angeles
2  Oliver   35      Chicago
PS C:\Users\USER\AppData\Local\Programs\Mi
```

Read and write data using pyspark :

```python
from pyspark.sql import SparkSession

# Initialize Spark session
spark = SparkSession.builder.appName("SparkExample").getOrCreate()
# Schema
schema = ["Name", "Age", "City"]
# Data
data = [
    ("Jacob", 30, "New York"),
    ("Lauren", 35, "Los Angeles"),
    ("Billy", 25, "Chicago")
]

# Create DataFrame
df = spark.createDataFrame(data, schema)

# Show result
#df.show()
# Write DataFrame to Parquet
df.write.parquet("dataa.parquet")
# Read the Parquet file
parquet_df = spark.read.parquet("dataa.parquet")

# Show the DataFrame
parquet_df.show()
```

```
+------+---+-----------+
|  Name|Age|       City|
+------+---+-----------+
|Lauren| 35|Los Angeles|
| Billy| 25|    Chicago|
| Jacob| 30|   New York|
+------+---+-----------+
```

Append data to the file and select specific columns:

[15]
0s

```python
import pyarrow.parquet as pq
import pyarrow as pa
import pandas as pd
# Load existing Parquet file
existing_table = pq.read_table("dataa.parquet")

# New data
new_data = pd.DataFrame({
    "Name": ["David", "Emma"],
    "Age": [40, 28],
    "City": ["San Francisco", "Seattle"]
})

# Convert new data to PyArrow table
new_table = pa.Table.from_pandas(new_data)

# Concatenate both tables
merged_table = pa.concat_tables([existing_table, new_table])

# Write back to Parquet file
pq.write_table(merged_table, "data1.parquet")
df = pd.read_parquet("data1.parquet", columns=["Name", "Age"])
print(df)
```

```
     Name  Age
0   Jacob   30
1  Lauren   35
2   Billy   25
3   David   40
4    Emma   28
```

Filters by rows:

[16]
0s

```python
import pyarrow.parquet as pq

# Read only rows where Age > 30
table = pq.read_table("data1.parquet", filters=[("Age", ">", 30)])

df = table.to_pandas()
print(df)
```

```
     Name  Age           City
0  Lauren   35    Los Angeles
1   David   40  San Francisco
```